# Supplement 1: Supporting content

## Appendix A. Theoretical derivations

### *A.1 Finite-$M$ Airy crosstalk: closed form and large-$M$ limit*

For $M$ uniformly spaced target modes in a cavity of finesse $\mathcal{F}$, the general-finesse form of the Airy circular sum used in the main text is

$$P_{\text{noise}}^{(M)}(\mathcal{F}) = \sum_{n=1}^{M-1} \frac{1}{1 + \left(\frac{2\mathcal{F}}{\pi}\right)^2 \sin^2\!\left(\frac{\pi n}{M}\right)}. \tag{S1}$$

With $a = (2\mathcal{F}/\pi)^2$, this sum can be evaluated in closed form using the standard trigonometric-sum identity

$$\sum_{n=0}^{M-1} \frac{1}{1 + a\sin^2(\pi n/M)} = \frac{M}{\sqrt{1+a}}\,\frac{1+q^M}{1-q^M}, \tag{S2}$$

This identity is valid for $a > 0$, with $q = (\sqrt{1+a}-1)/(\sqrt{1+a}+1)$ and $0 < q < 1$. The crosstalk sum in Eq. (S1) excludes the $n = 0$ contribution, which is unity. Subtracting this contribution yields the closed-form expression

$$P_{\text{noise}}^{(M)}(\mathcal{F}) = \frac{M}{\sqrt{1+a}}\,\frac{1+q^M}{1-q^M} - 1. \tag{S3}$$

The corresponding extinction ratio and sorting efficiency are

$$\mathrm{ER}_{\text{sum}}^{(M)} = 10\lg\frac{1}{P_{\text{noise}}^{(M)}}, \qquad \eta_{\text{sort}}^{(M)} = \frac{1}{1+P_{\text{noise}}^{(M)}}. \tag{S4}$$

**Capacity-boundary specialization and large-$M$ limit.** At the capacity-boundary finesse $\mathcal{F} = M\tau_0$, the sharpness parameter becomes $a_M = (2M\tau_0/\pi)^2$. As $M \to \infty$ at fixed $\tau_0$, $a_M \to \infty$ and $\sqrt{1+a_M} \sim 2M\tau_0/\pi$. Thus, the prefactor satisfies $M/\sqrt{1+a_M} \to \pi/(2\tau_0)$. For $q_M = q|_{a=a_M}$, the large-$M$ expansion gives $q_M \approx 1 - 2/\sqrt{a_M} = 1 - \pi/(M\tau_0)$, and therefore $(q_M)^M \to e^{-\pi/\tau_0}$. Substitution into Eq. (S3) yields

$$P_{\text{noise}}^{(\infty)}(\tau_0) = \frac{\pi}{2\tau_0}\,\frac{1+e^{-\pi/\tau_0}}{1-e^{-\pi/\tau_0}} - 1 = \frac{\pi}{2\tau_0}\coth\frac{\pi}{2\tau_0} - 1. \tag{S5}$$

At $\tau_0 = 3$, Eq. (S5) gives $P_{\text{noise}}^{(\infty)} \approx 0.0898$. The corresponding system extinction ratio and sorting efficiency are $\mathrm{ER}_{\text{sum}}^{(\infty)} \approx 10.47$ dB and $\eta_{\text{sort}}^{(\infty)} \approx 91.76\%$, respectively, consistent with Table 1 in the main text. The finite-$M$ Airy sum approaches this limit from below, so the large-$M$ expression provides a conservative upper bound on the crosstalk: $P_{\text{noise}}^{(M)} < P_{\text{noise}}^{(\infty)}$ for all finite $M$.

**Numerical evaluation at $M = 9$.** Evaluating Eq. (S3) at the threshold finesse $\mathcal{F}_{\min} = 27$ gives $\mathrm{ER}_{\text{sum}}^{(9)} = 10.53$ dB and $\eta_{\text{sort}}^{(9)} = 91.9\%$. This value lies within 0.06 dB of the conservative large-$M$ limit and corresponds to the $M = 9$ row in Table 2 of the main text. At the measured finesse $\mathcal{F} = 32.21$, the same expression gives $\mathrm{ER}_{\text{sum}}^{(9)} = 12.04$ dB, matching the finite-$M$ Airy reference used in the main text.

### *A.2 Optimal geometry for general target sets: rationality proof*

For a general target set $S = \{N_1, \ldots, N_M\}$, define the pairwise order-difference set as $D(S) = \{N_j - N_i : i < j\}$. The minimum folded spacing is

$$s_{\min}(k; S) = \min_{d \in D(S)} \|d\, k\|, \tag{S6}$$

where $\|x\| = \min_{n\in\mathbb{Z}} |x - n|$ denotes the distance from $x$ to its nearest integer. The design task is to find the value $k^*$ that maximizes $s_{\min}(k; S)$ over a rational search interval $I = [a, b]$.

For each difference $d \in D(S)$ and each integer $n$, the function $|dk - n|$ is a V-shaped piecewise-linear function of $k$ with vertex at $k = n/d$. Over the finite interval $I$, only finitely many integers $n$ are relevant. Therefore, $s_{\min}(k; S)$ is the pointwise minimum of finitely many such V-profiles and hence a piecewise-linear lower envelope.

On any linear segment of the lower envelope, the slope is nonzero. An interior maximum therefore cannot occur within such a segment; the global maximum must occur at a breakpoint or at an endpoint of $I$. The relevant breakpoints are of two types: (i) V-profile vertices $n/d$, which are rational with denominator at most $D_{\max} = \max D(S)$; and (ii) intersections of two V-profiles with slopes $+d_1$ and $-d_2$, which give $k = (n_1 + n_2)/(d_1 + d_2)$ and hence have denominator at most $d_1 + d_2 \le 2D_{\max}$. The endpoints $a$ and $b$ are rational by assumption. Thus, every candidate for the global maximum is rational, and the maximizing value $k^* = m/q$ has a denominator satisfying

$$q \le Q_I(S) = \max\big(\mathrm{den}(a),\ \mathrm{den}(b),\ 2D_{\max}\big), \tag{S7}$$

where $\mathrm{den}(\cdot)$ denotes the denominator of a rational number written in lowest terms. The continuous optimization therefore reduces to an exact comparison over finitely many rational candidates $(m, q)$ with $\gcd(m, q) = 1$ and $a \le m/q \le b$.

## Appendix B. Experimental calibration and boundary validation

### *B.1 Projection-arm calibration-correction procedure*

The mode-resolved projection arm consists of the second spatial light modulator (SLM2), a $4f$ relay lens system, and a single-mode fiber (SMF). SLM2 imprints conjugate complex-amplitude holograms onto the transmitted field [1] before fiber coupling. The power coupled into an SMF is proportional to the squared modal overlap between the field at the fiber plane and the fiber fundamental mode [2]:

$$P_i \propto \left| \int\!\!\int u_{\mathrm{f}}^*(\mathbf{r})\, E_i(\mathbf{r})\, d^2\mathbf{r} \right|^2. \tag{S8}$$

Here, $E_i(\mathbf{r})$ is the projected field for readout channel $i$ and $u_{\mathrm{f}}(\mathbf{r})$ is the fiber mode. The raw detected power therefore reflects the complete response of the projection arm, including channel-dependent first-order hologram throughput, finite spatial filtering, overlap with the fixed fiber mode, and residual projection crosstalk. Under the fixed projection-arm alignment used in the measurements, the overall response is largest for the $l = 1$ calibration channel. The diagonal entries are empirical responses of the aligned projection arm and are not expected to follow a monotonic sequence with mode index. Related holographic projection measurements of LG modes have also shown mode-dependent readout efficiencies [3, 4]. Accordingly, we independently characterize the projection arm and use the calibrated response in the simultaneous nine-mode measurement.

For each projection channel $i$ and prepared input mode $k$, we measure the readout-response matrix [5],

$$\mathbf{S} = \{S_{ik}\}, \qquad S_{ik} = \frac{P_{\mathrm{SMF}}(i \mid k)}{P_{\mathrm{trans}}(k)}. \tag{S9}$$

Here, $P_{\mathrm{SMF}}(i \mid k)$ denotes the detected power in projection channel $i$ when the input mode is prepared with azimuthal index $l = k$ and the cavity is locked to that mode. The denominator $P_{\mathrm{trans}}(k)$ is the transmitted reference power of the locked channel. This normalization places the calibration columns on the same transmitted-power scale and records the relative response of the sequential projection-arm readout.

Fig. S1 shows the readout-response matrix and the corresponding column-normalized raw projection matrix from the first of three independent experimental runs. Panel (b) visualizes the measured detected-power vectors after column normalization and before response correction; the inversion below uses the unnormalized vector $\mathbf{y}^{(j)}$. The same calibration-correction procedure is applied separately to each run before aggregation.

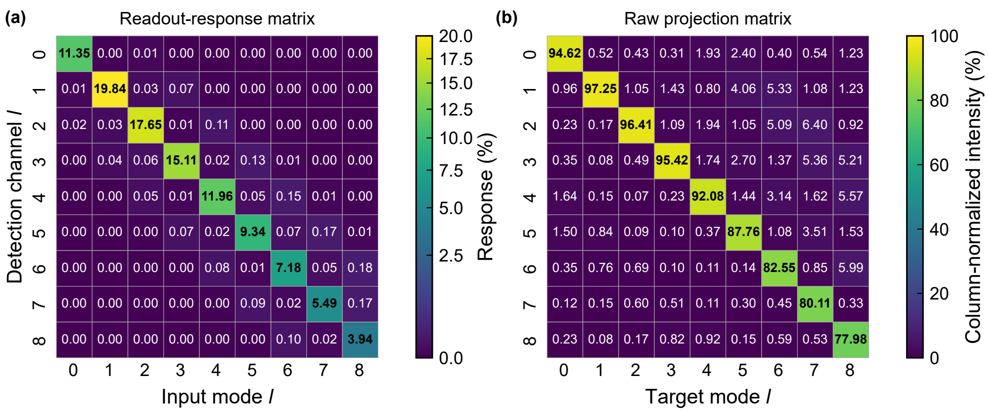


Fig. S1. Projection-arm readout response and raw projection data for Run 1. (a) Readout-response matrix $\mathbf{S}$. The diagonal responses span 3.94%–19.84%, the maximum off-diagonal response is 0.180%, and the condition number is $\kappa(\mathbf{S}) = 5.05$. (b) Column-normalized raw projection matrix measured for the simultaneous nine-mode input before response correction. Values in panel (b) are normalized for visualization; the inversion uses the raw detected-power vectors.

For the simultaneous nine-mode measurement at locking condition $j$, let $\mathbf{y}^{(j)}$ denote the raw detected-power vector across the nine projection channels and let $\mathbf{x}^{(j)}$ denote the corresponding cavity-output channel weights. Within each experimental run, $\mathbf{S}$ is the run-specific readout-response matrix. The measurement model is

$$\mathbf{y}^{(j)} = \mathbf{S}\,\mathbf{x}^{(j)}. \tag{S10}$$

The channel-weight vector is estimated by non-negative least-squares inversion [6]:

$$\hat{\mathbf{x}}^{(j)} = \arg\min_{\mathbf{x}\geq 0} \left\| \mathbf{S}\mathbf{x} - \mathbf{y}^{(j)} \right\|_2 . \tag{S11}$$

The estimated vectors are then normalized column-wise so that each column sums to unity. The sorting efficiencies and extinction ratios reported in the main text are calculated from the resulting response-corrected matrix. The three independently measured readout-response matrices have condition numbers from 4.99 to 5.43, so the inversion is well conditioned. The calibration and simultaneous nine-mode measurements use the same sequential single-channel projection-arm configuration, so $\mathbf{S}$ is applied under the readout conditions in which it is acquired.

### B.2 Design-point tolerance at the nine-mode boundary

Fig. S2 quantifies the robustness of the $M = 9$ consecutive design point against deviations $\Delta k$ from the optimum $k^* = 2/9$. Panel (a) plots the minimum resolvability $\tau_{\min} = \mathcal{F}\, s_{\min}$ as a function of $\Delta k$ for the threshold finesse $\mathcal{F}_{\min} = 27$ and the measured finesse $\mathcal{F} = 32.21$. At $\mathcal{F}_{\min}$, the threshold $\tau_{\min} \geq \tau_0 = 3$ is satisfied only at the exact optimum. The measured finesse opens a tolerance window $\Delta k \in [-3.6, +4.5] \times 10^{-3}$ within which $\tau_{\min}$ remains above $\tau_0$. Panel (b) shows the corresponding system extinction ratio $\mathrm{ER}_{\mathrm{sum}}$ at $\mathcal{F} = 32.21$, evaluated from the Airy closed form [Eq. (S3)]. At the design point $\Delta k = 0$, the mean and worst-mode $\mathrm{ER}_{\mathrm{sum}}$ both reach the exact finite-$M$ Airy prediction of 12.04 dB. Both values remain above the conservative lower bound of 10.47 dB set by the large-$M$ limit throughout the tolerance window.

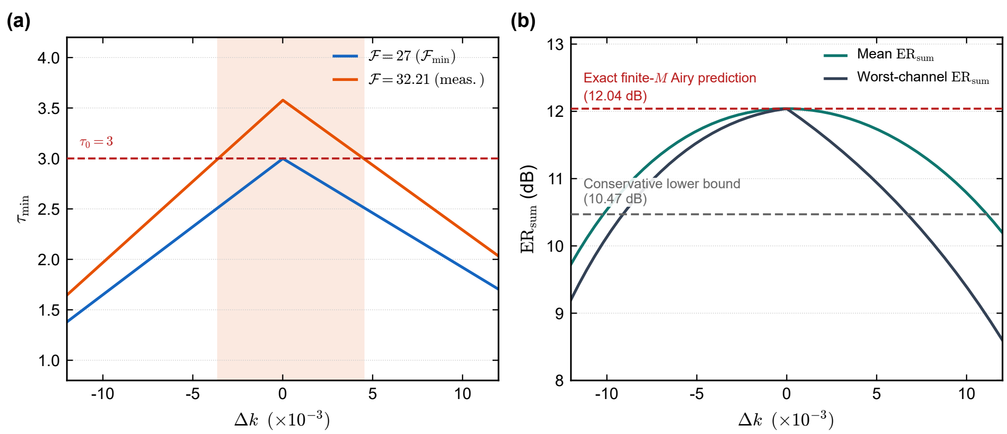


Fig. S2. Tolerance of the $M = 9$ consecutive design point. (a) Minimum resolvability $\tau_{\min}$ versus Gouy-step deviation $\Delta k$ at the threshold finesse $\mathcal{F}_{\min} = 27$ (blue) and the measured finesse $\mathcal{F} = 32.21$ (orange). The shaded region marks the tolerance window where $\tau_{\min} \geq \tau_0 = 3$. (b) System extinction ratio $\mathrm{ER}_{\mathrm{sum}}$ versus $\Delta k$ at the measured finesse, showing mean and worst-mode values. The horizontal dashed reference lines mark the conservative lower bound of 10.47 dB set by the large-$M$ limit and the exact finite-$M$ Airy prediction of 12.04 dB at the design point $k^* = 2/9$.

## Appendix C. Extension to general stable two-mirror cavities

The main text derives the folding model for a plane-concave cavity. The design analysis, however, depends on the cavity geometry only through the normalized Gouy step $k$. This section shows that a branch-independent effective step can be defined for a general stable two-mirror cavity. The folding positions, spacing criterion, crosstalk sums, and capacity bound therefore carry over without modification.

Consider a stable Fabry–Pérot cavity of length $L$ formed by two mirrors with radii of curvature $R_1$ and $R_2$. The standard $g$-parameters are

$$g_1 = 1 - \frac{L}{R_1}, \qquad g_2 = 1 - \frac{L}{R_2}, \tag{S12}$$

The stable regime considered here satisfies $0 < g_1 g_2 < 1$. The transverse-mode resonance frequencies are given by [7, 8]

$$\nu_{q,N} = \mathrm{FSR}\left(q + N\frac{\eta}{\pi}\right), \qquad \eta = \arccos\left(\sigma\sqrt{g_1 g_2}\right), \tag{S13}$$

where $\sigma = +1$ for the near-planar branch and $\sigma = -1$ for the near-concentric branch. On the folded circle, steps $k$ and $1-k$ give the same set of pairwise arc lengths. The branch-independent effective step can therefore be written as

$$k_{\mathrm{eff}} = \frac{1}{\pi}\arccos\left(\sqrt{g_1 g_2}\right) \in \left(0, \tfrac{1}{2}\right). \tag{S14}$$

With this replacement, the folded positions $\mathrm{pos}(N)$, pairwise spacings $s_{ij}$, and minimum spacing $s_{\min}$ are determined by the target orders and $k_{\mathrm{eff}}$, rather than by the cavity type itself. The same spacing criterion, crosstalk sums, and capacity bound $M \le \lfloor \mathcal{F}/\tau_0 \rfloor$ therefore remain applicable. Only the mapping from $k_{\mathrm{eff}}$ to the geometric parameters changes with cavity type.

**Plane-concave cavity.** Taking the planar limit $R_1 = \infty$ and denoting $R_2 = R$ gives $g_1 = 1$ and $g_2 = 1 - L/R$. Equation (S14) then reduces to $k_{\mathrm{eff}} = \pi^{-1} \arccos\big(\sqrt{1 - L/R}\big)$, recovering the main-text formula. The inverse mapping is $L/R = \sin^2(\pi k_{\mathrm{eff}})$.

**Symmetric double-concave cavity.** For $R_1 = R_2 = R$, the $g$-parameters satisfy $g_1 = g_2 = 1 - L/R$. Equation (S14) becomes

$$k_{\mathrm{eff}} = \frac{1}{\pi} \arccos\left( \left| 1 - \frac{L}{R} \right| \right). \tag{S15}$$

Solving for the geometry ratio yields

$$\frac{L}{R} = 1 \pm \cos(\pi k_{\mathrm{eff}}), \tag{S16}$$

where the minus sign corresponds to the near-planar branch ($0 < L/R < 1$) and the plus sign corresponds to the near-concentric branch ($1 < L/R < 2$). In this symmetric geometry, the waist lies at the cavity centre and has Rayleigh range $z_R = \frac{1}{2}\sqrt{L(2R - L)}$; by comparison, the plane-concave geometry has $z_R = \sqrt{L(R - L)}$. In the near-planar small-$L/R$ limit,

$$k_{\mathrm{pc}} \approx \frac{\sqrt{L/R}}{\pi}, \qquad k_{\mathrm{dc}} \approx \frac{\sqrt{2L/R}}{\pi} = \sqrt{2}\, k_{\mathrm{pc}}. \tag{S17}$$

Thus, at the same geometry ratio $L/R$, the near-planar double-concave cavity has a Gouy step larger than that of the plane-concave cavity by a factor approaching $\sqrt{2}$.

**General asymmetric two-mirror cavity.** For an asymmetric cavity with $R_1 \ne R_2$, the condition $g_1 g_2 = \cos^2(\pi k_{\mathrm{eff}})$ yields a quadratic equation for $L$:

$$L = \frac{R_1 + R_2 \pm \sqrt{(R_1 + R_2)^2 - 4R_1 R_2 \sin^2(\pi k_{\mathrm{eff}})}}{2}. \tag{S18}$$

The smaller and larger roots give the near-planar and near-concentric realizations, respectively. The plane-concave limit and the symmetric double-concave case recover the preceding mappings.

**Branch sensitivity.** The geometry-to-step derivative provides a local measure of how strongly a given coprime branch responds to fabrication tolerances. For the plane-concave cavity,

$$\left| \frac{dk}{d(L/R)} \right|_{\mathrm{pc}} = \frac{1}{2\pi \sqrt{(L/R)(1 - L/R)}}. \tag{S19}$$

This sensitivity is minimized at $L/R = 1/2$. For the symmetric double-concave cavity,

$$\left| \frac{dk}{d(L/R)} \right|_{\mathrm{dc}} = \frac{1}{\pi \sqrt{(L/R)(2 - L/R)}}. \tag{S20}$$

The minimum occurs at $L/R = 1$, corresponding to the confocal point. As a result, the most robust $M = 9$ branch shifts from $m = 2$ in the plane-concave geometry to $m = 4$ in the symmetric double-concave geometry. The optimal step condition $k^* = m/M$ and the capacity bound $M \le \lfloor \mathcal{F}/\tau_0 \rfloor$ remain unchanged; only the geometric realization and the robustness ranking change with cavity type.

## References


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